

Mayank Garg

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PROFESSIONAL SUMMARY

Software Development Engineer focused on scalable backend systems, microservices, and cloud architecture. Skilled in designing high-performance REST APIs (Node.js, FastAPI), AWS serverless pipelines, and secure cryptographic applications.

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, SQL

Web Development: React.js, Node.js, Express.js, FastAPI, Flask, RESTful APIs, HTML5, CSS3, Tailwind CSS

Databases: MongoDB, MySQL

Cloud& DevOps: AWS (Lambda, EC2, S3, SageMaker, CloudWatch, Auto Scaling), Docker, Git, GitHub, Linux, CI/CD

WORK EXPERIENCE

Project Intern – Cloud-Based IoT Data Analytics

Jun 2024 – Jul 2024

IBM

Remote

- Architected a **serverless data pipeline** using **AWS Lambda**, **IoT Core** (MQTT), and **Amazon SNS**, ingesting real-time sensor telemetry into **DynamoDB** and **RDS (MySQL)** with automated threshold-based SMS and email alerts.
- Built a farmer-facing monitoring dashboard using **React.js** and **Chart.js** with real-time soil moisture and temperature visualizations, achieving sensor-to-dashboard latency of **under 2 seconds**.
- Configured **AWS SageMaker** to train a Random Forest model for predictive irrigation scheduling, achieving a projected **20% reduction in water consumption** during field testing.

Research Intern

Jun 2022 – Jul 2022

GiftAble Foundation

Remote

- Analyzed **20 years** of district-level census data for Bahraich, Uttar Pradesh, generating analytical snapshots on women's and disability welfare; contacted government departments to supplement **2011 Census** data with current figures, identifying **5+ strategic intervention areas**.
- Built a national database of **500+ college placement cells** (emails, contacts) across **7 states** in MS Excel, streamlining recruiter outreach for the NGO's partnership program across India.

PROJECTS

CloudSentry – Autonomous AWS Cost Optimizer | React, Node.js, FastAPI, Prophet, AWS SDK, Docker, MongoDB

- Designed a **four-tier microservices architecture** orchestrated with **Docker Compose** (React UI, Express API, Python ML engine, MongoDB), enabling predictive auto-scaling of AWS EC2 instances.
- Built a **Prophet** time-series forecasting engine (FastAPI) ingesting CloudWatch CPU metrics via cron jobs, predicting traffic surges **60 minutes ahead** and triggering scale-up/down via **AWS Auto Scaling API**.
- Implemented **JWT authentication**, **AES-256** credential encryption, Zod validation, Helmet headers, rate limiting, and **AWS Pricing API** integration for real-time per-instance cost tracking.

Decentralized Voting System | React, Vite, Node.js, Express, MongoDB, RSA-OAEP, SHA-256

- Engineered an end-to-end encrypted ballot system with **client-side RSA-OAEP encryption**, ensuring zero plaintext exposure of votes on the server, and a tamper-evident **SHA-256 hash-chained** cryptographic ledger.
- Built a one-time voter pass authentication system with **double-spend protection** and a real-time chain integrity verification API validating the hash linkage of each block in MongoDB.

SecurePixel – Layered Chaos Theory & Steganography Engine | Python, Flask, OpenCV, NumPy, MongoDB

- Engineered a two-layer image cryptography system using vectorized **Henon map** iterations (NumPy) for pixel scrambling and **2-bit LSB steganography** with 32-bit dynamic file headers, achieving a PSNR of **30+ dB**.
- Secured with **AES-256 (Fernet)** encryption for initialization vectors at rest, **CSRF protection** (Flask-WTF), rate limiting (Flask-Limiter), and an authenticated MongoDB ledger tracking all encryption operations.

EDUCATION

University of Petroleum and Energy Studies (UPES)

Aug 2021 – May 2025

B.Tech. (Hons.) Computer Science Engineering | CGPA: 7.66

Dehradun, India

LEADERSHIP EXPERIENCE

Associate Technical Head

May 2023 – May 2024

UPES Hypervision

Dehradun, India

- Led a team of **8 members**, coordinating the planning and execution of **6+ technical events**, streamlining workflows and ensuring timely delivery of high-impact club initiatives.

CERTIFICATIONS & ACHIEVEMENTS

AWS Certified Developer – Associate

Sept 2024

3rd Place – UPES Hypervision Code Hustle Hackathon (Automated YouTube Video Tags Updater)

Finalist – 48-Hour UPES CSI Hackathon (Image Encryption System)